RIGHT DERIVED FUNCTORS VANISH IN NEGATIVE DEGREES BELOW THE COHOMOLOGICAL SUPPORT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5727-5759, following the statement at 5713-5725.

Editorial note. No omissions: the summand argument behind the final 'It follows that' of part (1) is one step (F(L) concentrated in degrees >= a; direct summands inherit vanishing).

Allowed standard prerequisites (editorial).

- Cofinal restriction of essentially constant systems. Categories, Lemma categories-lemma-cofinal-essentially-constant (cross-chapter, cited around line 5734). Lets RF be computed using only quasi-isomorphisms with target concentrated in degrees >= a.
- Truncation triangle. remark-truncation-distinguished-triangle (same chapter, cited around line 5744). Provides the triangle tau_{<=a}K -> K -> tau_{>=a+1}K used in part (2).
- RF defined on two of three. lemma-2-out-of-3-defined (carded L26162, cited around line 5746). Extends definedness of RF along the truncation triangle.

Source reference. Original reference 0A1S: focused-seeds/geometry/S9-der-negative-vanishing/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05SC: sources/stacks-derived/pinned-original-source.tex, source lines 4896–4896. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 08J5: sources/stacks-derived/pinned-original-source.tex, source lines 4293–4293. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor between abelian categories. Let $K^\bullet$ be a complex of $\mathcal{A}$ and $a \in \mathbf{Z}$.*

(1) *If $H^i(K^\bullet) = 0$ for all $i < a$ and RF is defined at $K^\bullet$, then $H^i(RF(K^\bullet)) = 0$ for all $i < a$.*

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *If RF is defined at $K^\bullet$ and $\tau_{\leq a}K^\bullet$, then $H^i(RF(\tau_{\leq a}K^\bullet)) = H^i(RF(K^\bullet))$ for all $i \leq a$.*

Proof. Assume $K^\bullet$ satisfies the assumptions of (1). Let $s : K^\bullet \to L^\bullet$ be any quasi-isomorphism. Then it is also true that $K^\bullet \to \tau_{\geq a}L^\bullet$ is a quasi-isomorphism by our assumption on $K^\bullet$. Hence in the category $K^\bullet/\mathrm{Qis}^+(\mathcal{A})$ the quasi-isomorphisms $s : K^\bullet \to L^\bullet$ with $L^n = 0$ for $n < a$ are cofinal. From Categories, Lemma 0A1S we deduce that RF is the value of the essentially constant ind-object $F(L^\bullet)$ for these s. This means that $\mathrm{id} : RF(K^\bullet) \to RF(K^\bullet)$ factors through $F(L^\bullet)$ for some complex $L^\bullet$ with $L^n = 0$ for $n < a$. It follows that $H^i(RF(K^\bullet)) = 0$ for $i < a$.

To prove (2) we use the distinguished triangle

$$\tau_{\leq a}K^\bullet \to K^\bullet \to \tau_{\geq a+1}K^\bullet \to (\tau_{\leq a}K^\bullet)[1]$$

of Remark 08J5 to conclude via Lemma 05SC that RF is defined at $\tau_{\geq a+1}K^\bullet$ as well and that we have a distinguished triangle

$$RF(\tau_{\leq a}K^\bullet) \to RF(K^\bullet) \to RF(\tau_{\geq a+1}K^\bullet) \to RF(\tau_{\leq a}K^\bullet)[1]$$

in $D(\mathcal{B})$. By part (1) we see that $RF(\tau_{\geq a+1}K^\bullet)$ has vanishing cohomology in degrees $< a+1$. The long exact cohomology sequence of this distinguished triangle then shows what we want. □